

Wipro Capstone Project — Planning Document

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Project Overview

Website	Practice Software Testing (practicesoftwaretesting.com)	Framework	Playwright v1.4x+ JavaScript Node.js
API	Free REST API — api.practicesoftwaretesting.com (Swagger)	Pattern	Page Object Model (POM) + Custom Fixtures
Test Cases	120+ (8 services × 15 each)	Reporting	Allure Report + Playwright HTML Report
AI Tools	GitHub Copilot + MCP Integration	CI/CD	Jenkins (local + headless execution)

Why Practice Software Testing?

Looks Like a Real Store	Free API — No Cost
Angular e-commerce frontend mimicking a real tool shop — professional appearance	Full Swagger REST API at no cost — covers auth, products, cart, orders, users

Services & Test Case Breakdown (8 × 15 = 120 Test Cases)

#	Service	Key Test Scenarios	Cases
1	Authentication	Valid login, invalid login, logout, registration, password reset, session persistence, token expiry, locked account, guest access, remember me, concurrent sessions, brute force, empty fields, SQL injection attempt, redirect after login	15
2	Product Catalog	Search by name, filter by category, sort by price (asc/desc), product detail page, image display, out-of-stock label, pagination, breadcrumb nav, related products, price accuracy, URL routing, add to favourites, no-results state, category count, page load	15
3	Cart	Add item, remove item, update quantity, empty cart state, price total, apply coupon, cart persistence after login, max quantity limit, cart badge update, duplicate item handling, cart summary accuracy, continue shopping, tax display, checkout redirect, guest cart	15
4	User Profile	Update name, update email, change password, view order history, form validation errors, required field check, phone number update, success confirmation, cancel changes, password strength rule, email format validation, duplicate email, session after update, account info display, delete account	15
5	Address & Shipping	Add new address, edit existing address, delete address, set default address, multiple saved addresses, shipping method selection, delivery estimate display, address form validation, city/state fields, pincode validation, billing vs shipping toggle, free shipping threshold, address limit, international address, autofill	15
6	Payment & Checkout	Full checkout flow, order confirmation page, invalid card input, order total calculation, tax calculation, discount code applied, payment loading state, order ID generated, back button during checkout, checkout progress steps, empty cart redirect, address selection at checkout, terms acceptance, place order button, receipt display	15
7	Customer Support	Contact form submission, required field validation, success message, invalid email format, message character limit, form reset, category dropdown, subject line validation, response time info displayed, blank form submit, special characters in message, multiple submissions, field placeholder text, mobile view form, FAQ section	15
8	API (Internal)	GET /products, GET /product/{id}, POST /login, POST /register, PUT /user/{id}, GET /cart, POST /cart, DELETE /cart/{id}, GET /orders, GET /categories, 401 unauthorized check, 404 not found check, response schema validation, API response time under 2s, invalid token handling	15

Implementation Milestones

Milestone	Deliverable
M1 — Setup	Node.js workspace, playwright.config.js profiles, package.json scripts: test:headed test:parallel test:debug test:report
M2 — Locators	getByRole() semantic locators, form automation, soft assertions (expect.soft), data-driven inputs via JSON files
M3 — Advanced UI	Table automation, Shadow DOM piercing, iframe handling via frameLocator(), drag-drop, file upload, dialog & multi-tab management
M4 — Reporting	Allure Report with trends + Playwright HTML Report, failure screenshots/videos, trace viewer, visual regression with field masking
M5 — API & Mocking	API auth bypass via Playwright request context, page.route() for backend mocking, edge-case status code simulation (500, 404, 401)
M6 — AI & MCP	GitHub Copilot for test generation & locator refactoring, MCP host config exposing DOM state to LLM, prompt files documented in repo

